

Manatees and Sand Tiger Sharks Use Gas to Stay Afloat

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Flatulence, the release of gas from the body's lower opening, might seem like an unlikely survival tool. Yet some aquatic animals use the gas inside their bodies to help control their position in the water.

Florida manatees provide one example. According to data scientist and animal-**flatulence** author Nick Caruso, they can store gas in their guts. The gas may come from swallowed air or from digestion. Because gas is less dense than water, it increases **buoyancy**: the tendency to float rather than sink.

Manatees mainly eat seagrass in Florida's shallow waters. These plants are difficult to digest, so the animals rely on a large **hindgut**, the rear part of the digestive tract. **Microbes** living there help break down the food, producing gas in the process.

The animals also have strong abdominal muscles and a powerful **diaphragm**, the muscle involved in breathing. These muscles can compress gas in the intestines and force some out. Releasing gas can help a manatee dive deeper. What people might dismiss as an awkward bodily function can therefore play a practical role underwater.

Sand tiger sharks use air for **buoyancy** too, but their method differs. Many bony fish have a swim bladder, an organ containing gas that helps them float. Sharks and rays lack both bones and swim bladders. Sharks' oily livers help support them because oil is lighter than water, although many sharks still keep swimming to avoid sinking.

Marine conservation biologist David Shiffman explains that sand tiger sharks can sometimes hover instead. They gulp air at the surface and use it to help them stay in the water column. Shiffman says these sharks have been observed expelling air through their **cloaca**.

A **cloaca** is an opening used for waste and reproduction. Birds, reptiles, amphibians, sharks and some fish have one. Most mammals, by contrast, have separate routes for urine and feces. These anatomical differences mean that gas does not leave every animal in exactly the same way.

Some turtles have another unusual use for a **cloaca** that is separate from **flatulence**. Zoologist Dani Rabaiotti explains that they pump water into it, allowing oxygen from the water to enter their blood. Australia's white-throated snapping turtle obtains 48 percent of its oxygen this way when in water.

These examples show why studying animal bodies can turn a funny question into useful biology. Gas storage, gas release and oxygen uptake are different processes, but each can help explain how particular animals function underwater.

Word Treasure

Flatulence -- The release of gas through the body's lower opening.

buoyancy -- The tendency to float rather than sink.

hindgut -- The rear part of the digestive tract.

Microbes -- Tiny living things that help manatees break down food.

diaphragm -- A muscle involved in breathing.

cloaca -- An opening used for waste and reproduction in some animals.

Background you need

Gas is less dense than water, which helps explain its role in buoyancy.

Many bony fish have swim bladders; sharks and rays do not.

Oxygen uptake through a turtle's cloaca differs from releasing gas through that opening.

Break it down

Introduces flatulence and the role of gas in controlling position underwater.

Explains manatee gas storage, digestion and muscle-controlled gas release.

Compares bony fish with sharks and explains how sand tiger sharks use swallowed air.

Defines a cloaca and describes oxygen uptake in some turtles.

Connects the examples to the study of how animal bodies function.

Why it matters

The article connects buoyancy, digestion and oxygen uptake to concrete animal examples, showing how different body processes help animals function underwater.

Test what you remember

1. According to Nick Caruso, where can gas in a manatee's gut come from?
 - (A) Blood or surrounding sand
 - (B) Seawater or liver oil
 - (C) Skin or breathing muscles
 - (D) Swallowed air or digestion
2. What happens as microbes break down food in a manatee's hindgut?
 - (A) They store liver oil.
 - (B) They form a swim bladder.
 - (C) They remove oxygen.
 - (D) They produce gas.
3. How can a manatee's strong muscles help it dive deeper?
 - (A) They squeeze some gas out.
 - (B) They pump water into its blood.
 - (C) They move oil into its gut.
 - (D) They pull air into its lungs.
4. Which structure helps many bony fish float but is absent in sharks?
 - (A) An oily liver
 - (B) A swim bladder
 - (C) An abdominal muscle
 - (D) A digestive tract
5. What does Shiffman say sand tiger sharks have been observed doing?
 - (A) Expelling air through their cloacas
 - (B) Making gas by digesting seagrass
 - (C) Taking oxygen through their cloacas
 - (D) Moving air into their swim bladders
6. When in water, what share of its oxygen does Australia's white-throated snapping turtle obtain through its cloaca?
 - (A) About one quarter
 - (B) About three quarters
 - (C) All of its oxygen
 - (D) Exactly 48 percent

Answer key (try the quiz first!): 1: D 2: D 3: A 4: B 5: A 6: D

1. Nick Caruso: He explains that Florida manatees store gas from swallowed air or digestion and that gas increases their buoyancy.
2. David Shiffman: He explains that sand tiger sharks gulp air to hover and have been observed expelling air through their cloacas.
3. Dani Rabaiotti: She explains that some turtles pump water into their cloacas, allowing oxygen from the water to enter their blood.

MY THOUGHTS (at least 20 words):